

megaSoC Datasheet

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Chapter 1

Introduction

1.1 Summary

Chapter 2

System

2.1 Bus Interconnect

2.2 Address Map

2.2.1 Summary

Region	Start Address	End Address
Boot-Rom	0x00000000	0x0000FFFF
FLASH	0x00400000	0x007FFFFFFF
SRAM	0x00800000	0x0080FFFF
FLASH CTRL	0x01000000	0x0100FFFF
DMA CTRL	0x01010000	0x01011FFF
GIC	0x01100000	0x01107FFF
PCK CTRL	0x01200000	0x01207FFF
PERIPHERAL	0x40000000	0x5FFFFFFF
DEBUG	0x60000000	0x7FFFFFFF
DRAM	0x80000000	0xFFFFFFFF

2.2.2 Peripheral Region

Below are the address regions for the System IO/Peripherals, detailed address maps for each peripheral are in chapter 3

Region	Start Address	End Address
Timer 0	0x40000000	0x40000FFF
Timer 1	0x40001000	0x40001FFF
Dual Timer	0x40002000	0x40002FFF
USRT 0	0x40003000	0x40003FFF
USRT 1	0x40004000	0x40004FFF
UART 0	0x40005000	0x40005FFF
UART 1	0x40006000	0x40006FFF
Watchdog	0x40007000	0x40007FFF
RTC	0x40008000	0x40008FFF
SPI	0x40009000	0x40009FFF

2.3

Chapter 3

Peripherals

Chapter 4

Pin Outs

MegaSoC has 2 GPIO ports, P0 and P1. P0 is a completely general purpose with no alternative functionality. P1 can be used for some peripherals, UART0, UART1, SPI and EXTIO. The mapping for P1 is as follows

P1 Pin	Alt Function
0	UART0 Rx
1	UART0 Tx
2	UART1 Rx
3	UART1 Tx
4	SPI SSn
5	SPI SCLK
6	SPI MOSI
7	SPI MISO
8	EXTIO Data 0
9	EXTIO Data 1
10	EXTIO Data 2
11	EXTIO Data 3
12	EXTIO Req 1
13	EXTIO Req 2
14	EXTIO Ack
15	N/A

Chapter 5

Recommended Testboard